

.NET TECHNICAL CONCEPTS REPORT

1. .NET Versions

.NET has evolved from the Windows-only .NET Framework into a unified, open-source ecosystem starting with .NET Core.

Modern releases like .NET 6, 7, 8, and 9 deliver high performance and cross-platform capabilities under a single framework umbrella.

Each new iteration introduces optimized runtime performance, updated C# language features, and long-term support (LTS) stability.

2. Namespace

A namespace provides a logical container to organize related classes and types within a project structure.

It prevents naming collisions by creating distinct scopes for identifier names across different libraries.

Developers use the using directive in C# to clean up code and access namespaces effortlessly.

3. .NET Core

.NET Core is a complete redesign of the legacy framework built specifically for modern cloud and cross-platform development.

It runs seamlessly on Windows, macOS, and Linux using a fast, modular runtime architecture.

Its lightweight design enables side-by-side deployment and efficient microservices containerization using Docker.

4. Solution (.sln)

A Visual Studio Solution acts as an overarching container used to manage one or multiple related projects together.

It maintains project dependencies, build configurations, and global source control settings in a single file.

This logical grouping simplifies building, debugging, and maintaining enterprise application suites.